

Chapter 8

Dental Biofilms

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Introduction

Humans have evolved to have an intimate and dynamic relationship with microorganisms; this includes those that make up the resident microbiota of all environmentally-exposed surfaces of the body as well as those that cause disease. Contemporary studies are demonstrating that the relationship with the resident microbiota is highly interactive, and makes a major contribution to the health of the host. This relationship is dynamic and fragile, and a number of intrinsic and extrinsic factors can perturb this exquisite balance. An understanding of the relationship between host and oral microbiota is critical for the effective clinical management of dental patients both during health and when treating disease.

A remarkable statistic is that the human body is estimated to be composed of $>10^{14}$ cells, of which only 10% are mammalian (Sanders & Sanders 1984; Wilson 2005). The majority are the microorganisms that make up the resident microbiotas that colonize all accessible surfaces of the body and which, as will be discussed later, confer significant benefits to the host. These microbiotas have a diverse composition, and function as interactive microbial communities in which their combined properties are greater than the sum of the activities of the constituent species (see later). The microbiotas of the skin, mouth, digestive and reproductive tracts are distinct from each other despite the frequent transfer of organisms between these sites; their characteristic composition is a consequence of significant differences in the biologic and physical properties of each habitat (Wilson, 2005).

These properties determine which microorganisms are able to colonize successfully, and which will predominate or be only a minor component of the established microbial community.

The mouth as a microbial habitat

The mouth is similar to other habitats within the body in having a characteristic microbial community that provides benefits for the host. The mouth is warm and moist, and is able to support the growth of a wide range of microorganisms, including viruses, mycoplasma, bacteria, Archaea, fungi, and protozoa (Wilson 2005; Marsh & Martin 2009). These microorganisms colonize mucosal and dental surfaces in the mouth to form three-dimensional, structurally-organized multispecies communities that are termed biofilms. The biofilms that form on teeth are referred to as dental plaque. In general, desquamation ensures that the microbial load on mucosal surfaces is kept relatively low. In contrast, the mouth is a unique site in the body in that it provides non-shedding surfaces (teeth, dentures) for microbial colonization. This can result in the accumulation of large numbers of microorganisms, particularly at stagnant and hard-to-reach sites, unless patients practice effective oral hygiene. The main focus of this chapter is to describe the properties of biofilms that develop on teeth (dental biofilms).

A number of environmental factors will influence the distribution and metabolic activity of the resident oral microbiota (Fig. 8-1a) (Marsh & Devine 2011).

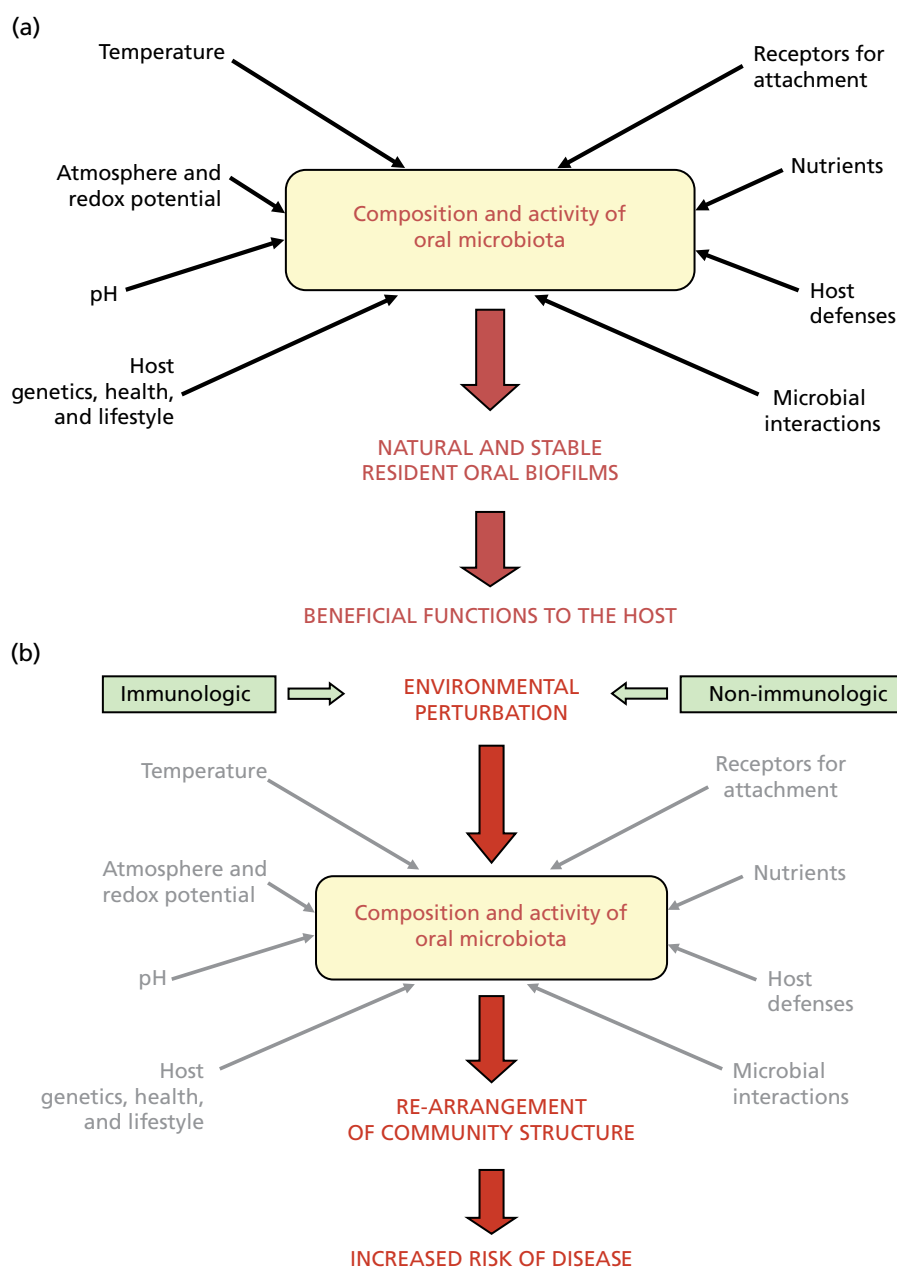


Fig. 8-1 Host factors that influence the microbial composition, activity, and stability of the resident oral microbiota. (a) A number of host factors help to determine the composition and activity of the natural oral microbiota, which provides benefits to the host. (b). Perturbation in a key environmental factor can disrupt the natural stability (microbial homeostasis) of the resident microbiota at a site and result in a rearrangement of the composition and activity of the resident microbial community; such a change might predispose the site to disease. (Source: Adapted from Marsh & Devine 2011, from John Wiley & Sons.)

The mouth is maintained at a temperature of around 35–37°C, which is suitable for the growth of a broad range of microbes. Temperature does increase at subgingival sites during inflammation, and this can alter bacterial gene expression, which in turn can alter the competitiveness of bacteria within the microbial community, and favor growth and protease activity of some putative periodontal pathogens. Although the mouth is overtly aerobic, the majority of oral bacteria are facultatively or obligately anaerobic. The distribution of these anaerobes in the mouth is generally related to the redox potential (Eh), the measure of the degree of oxidation–reduction at a site. The gingival crevice has the lowest Eh in the healthy mouth, and

harbors the largest numbers of obligate anaerobes (Kenney & Ash 1969). As oral bacteria exist as members of microbial communities, some anaerobic species survive in more aerobic habitats by existing in close partnership with oxygen-consuming species. Bacterial metabolism in mature oral biofilms results in sharp gradients of oxygen and Eh, thereby generating a mosaic of microenvironments suitable for the growth of bacteria with a range of oxygen tolerances. Many oral anaerobes also express a range of enzymes whose function is to scavenge low levels of oxygen in the environment to enable them to survive.

In the mouth, pH is a major determinant of bacterial distribution and metabolism. The buffering activity

of saliva plays a major role in maintaining the intraoral pH at around neutrality, which again is suitable for the growth of members of the resident oral microbiota. Changes in environmental pH frequently occur, and when they do they drive major shifts in the proportions of bacteria within dental plaque biofilms. After sugar consumption, the pH in plaque can fall rapidly to below 5.0 by the production of acidic fermentation products (Marsh & Martin 2009). Depending on the frequency of sugar intake, the bacteria in plaque will be exposed to varying challenges of low pH. Many of the predominant plaque bacteria that are associated with healthy sites can tolerate brief conditions of low pH, but are inhibited or killed by more frequent or prolonged exposures to acidic conditions (Svensater *et al.* 1997). This can result in the enrichment of acid-tolerant (aciduric) species, especially mutans streptococci, bifidobacteria, and lactobacilli, which are normally absent or only minor components in dental plaque at healthy sites. Such shifts in the bacterial composition of plaque predispose a surface to dental caries. The pH of the healthy gingival crevice is approximately 6.9, but this rises to between pH 7.2 and 7.4 during inflammation, with a few patients having pockets with a mean pH of around 7.8 (Eggert *et al.* 1991). Inflammation results in the increased flow of gingival crevicular fluid (GCF) in the subgingival habitat, and the rise in pH is a consequence of increased bacterial proteolysis of host proteins and glycoproteins in GCF. Even a small change in pH can alter the growth rate and pattern of gene expression in subgingival bacteria, and increase the competitiveness of some of the putative Gram-negative anaerobic pathogens at the expense of species associated with periodontal health (McDermid *et al.* 1988).

Saliva and GCF also have a major influence on bacterial distribution because they provide an array of host molecules that are potential nutrients for microorganisms. Primary nutrients, such as amino acids, proteins, and glycoproteins, are obtained from saliva and GCF; diet has only a minor role on the resident microbiota, mainly via the change in pH from sugar catabolism, as discussed earlier. The metabolism of complex host molecules requires the sequential or concerted action of consortia of bacteria (see later) in order to achieve their complete breakdown (ter Steeg & van der Hoeven 1989; Homer & Beighton 1992a, b; Bradshaw *et al.* 1994; Palmer *et al.* 2006; Periasamy & Kolenbrander 2009).

The mouth is richly endowed with components of both the innate (e.g. lysozyme, lactoferrin, sialoperoxidase, host defense peptides, neutrophils, etc.) and adaptive (secretory IgA, IgG, etc.) immune response (Marsh & Martin 2009). Complement, which bridges both the innate and adaptive immune responses, is also present. An area of considerable research activity at present concerns the relationship between the resident microbiota at any site and the host defenses, and how these microbial communities persist without

triggering an undesirable and unwanted detrimental host response, while the host retains the ability to respond to a genuine microbial challenge. Evidence is accumulating that some members of the resident oral microbiota are involved in active cross-talk with the host to down-regulate potential pro-inflammatory responses (Hasegawa *et al.* 2007; Cosseau *et al.* 2008).

The lifestyle of an individual can affect the distribution and metabolism of the oral microbiota (Marsh & Devine 2011). The impact of a diet with a high frequency of intake of fermentable carbohydrates has been discussed already. Smoking may select for potential periodontal pathogens in dental biofilms, and individuals with diabetes have a higher frequency of certain Gram-negative periodontal pathogens in plaque. The composition of the oral microbiota can also change with age as a consequence of a number of host-related events, including tooth eruption in early life or the waning of the immune response in old age. The influence of female hormones in GCF during pregnancy on the prevalence of some periodontal pathogens is controversial (Adriaens *et al.* 2009), although correlations have been found between maternal hormone levels and the increased proportions of black pigmented anaerobes, such as *Porphyromonas gingivalis* and *Prevotella intermedia*, in dental biofilms (Carrillo-de-Albornoz *et al.* 2010).

In general, once established, the microbial composition of the biofilm at a site remains stable over time, unless a major perturbation occurs in a key environmental determinant, such as a major change in diet or an alteration to the immune status of the host. Such perturbations can drive shifts in the balance of the microbiota, which can increase the risk of disease (Fig. 8-1b).

Significance of a biofilm and community lifestyle for microorganisms

The vast majority of microorganisms in nature, including those in the mouth, are found attached to surfaces as biofilms. Biofilms have been defined as matrix-embedded microbial populations, adherent to each other and/or to surfaces or interfaces (Costerton *et al.* 1995). The ability to attach to and be retained at a surface is a fundamental survival strategy for most prokaryotic organisms. Microorganisms would be lost from the mouth if they did not firmly attach to an oral surface and form a biofilm. The microbes found in multispecies biofilms are not randomly distributed but are spatially and functionally organized, and many natural biofilms have a highly diverse microbiota.

There would be little scientific or clinical interest if the properties of biofilms were simply those of planktonic (liquid culture) cells attached to a surface, and if the properties of microbial communities were merely the sum of those of the constituent species. However, studies have established that bacterial gene expression can alter markedly when cells form

a biofilm, resulting in many organisms having a radically different phenotype following attachment to a surface (Marsh 2005). Furthermore, the binding of bacteria to specific host receptors can also trigger significant changes in patterns of host cell gene expression. Most natural biofilms contain multiple species that engage in a wide range of physical, metabolic, and molecular interactions, and are termed microbial communities. This community lifestyle provides several potential benefits to the participating organisms (Caldwell *et al.* 1997; Shapiro 1998; Marsh & Bowden 2000) including:

- *Broader habitat range for growth.* For example, the metabolism of early colonizers alters the local environment, making conditions suitable for attachment and growth of later (and sometimes more fastidious) species.
- *Increased metabolic diversity and efficiency.* Molecules that are normally recalcitrant to catabolism by individual organisms can often be broken down by microbial consortia.
- *Enhanced tolerance of environmental stress, antimicrobial agents, and the host defenses.* Neighboring cells of a different species can produce neutralizing enzymes (β -lactamase, IgA protease, catalase, etc.) that protect inherently susceptible organisms from inhibitors (Brook 1989). Horizontal gene transfer is also more efficient in multispecies biofilms (Molin & Tolker-Nielsen 2003; Wilson & Salyers 2003). Microbial communities can also give physical protection from phagocytosis to cells deep within a spatially-organized consortium (Costerton *et al.* 1987; Fux *et al.* 2005).
- *Enhanced ability to cause disease.* Abscesses are examples of polymicrobial infections whereby organisms that individually cannot cause disease are able to do so when they are present as a consortium (pathogenic synergism) (van Steenberg *et al.* 1984).

Thus, microbial communities display emergent properties, in other words the properties of the community are more than the sum of its component populations.

An important clinical consequence of both the structural and functional organization of multispecies biofilms is their reduced susceptibility to antimicrobial agents (Gilbert *et al.* 1997; Ceri *et al.* 1999; Stewart & Costerton, 2001; Gilbert *et al.* 2002). Conventionally, the sensitivity of bacteria to antimicrobial agents is determined by growing them in liquid culture and measuring the minimum inhibitory concentration (MIC) or minimum bactericidal concentration (MBC). The MIC of an organism growing on a surface can be from 2- to 1000-fold greater than that of the same cells grown planktonically (Stewart & Costerton 2001), with older biofilms being most recalcitrant.

The mechanisms behind the increased tolerance of biofilms to antimicrobial agents are still the subject of much research (Stewart & Costerton 2001; Gilbert *et al.* 2002). Resistance conventionally develops due

to mutations affecting the drug target, to the presence of efflux pumps or to the production of modifying enzymes, etc., but even sensitive bacteria become less susceptible when growing on a surface. The structure of a biofilm may restrict the penetration of the antimicrobial agent; charged inhibitors can bind to oppositely-charged polymers that make up the biofilm matrix (diffusion–reaction theory). The agent may also adsorb to and inhibit the organisms at the surface of the biofilm, leaving cells in the depths of the biofilm relatively unaffected. The matrix in biofilms can also bind and retain neutralizing enzymes (e.g. β -lactamase) at concentrations that could inactivate an antibiotic or inhibitor (Allison 2003). As stated earlier, bacteria growing on a surface display a novel phenotype, and this can result in a reduced sensitivity to inhibitors, because the drug target may be modified or not expressed, or the organism may use alternative metabolic strategies. Bacteria grow only slowly under nutrient-depleted conditions in an established biofilm and, as a consequence, are much less susceptible than faster dividing cells. In addition, it has also been proposed that the environment in the depths of a biofilm may be unfavorable for the optimal action of some drugs (Gilbert *et al.* 2002). One hypothesis suggests that the increased tolerance of some biofilms to antibiotics is due largely to the presence of a subpopulation of “persister” organisms that are specialized survivor cells (Keren *et al.* 2004).

The most diverse collections of oral microorganisms are found in the biofilms on teeth (dental plaque) (Aas *et al.* 2005; Marsh & Martin 2009; Papaioannou *et al.* 2009; Dewhirst *et al.* 2010). It is now very common to see the term “dental biofilm” used instead of the original descriptor “dental plaque”. This does not mean that the original work performed on “dental plaque” is now invalid or irrelevant; rather it emphasizes the point that the broader principles derived from work on biofilms from across the spectrum of microbiologic habitats are directly relevant to dental biofilms, and *vice versa*.

Formation of dental biofilms

Dental biofilms form via an ordered sequence of events, resulting in a structurally- and functionally-organized, species-rich microbial biofilm (Socransky & Haffajee 2002; Marsh 2005; Kolenbrander *et al.* 2006; Marsh *et al.* 2011) (Fig. 8-2). The distinct stages in dental biofilm formation include:

1. Adsorption of a conditioning film (acquired pellicle)
2. Reversible adhesion between the microbial cell surface and the conditioning film
3. More permanent attachment involving interactions between specific molecules on the microbial cell surface (adhesins) and complementary molecules (receptors) present in the conditioning film
4. Co-adhesion, in which secondary colonizers adhere to receptors on already attached bacteria

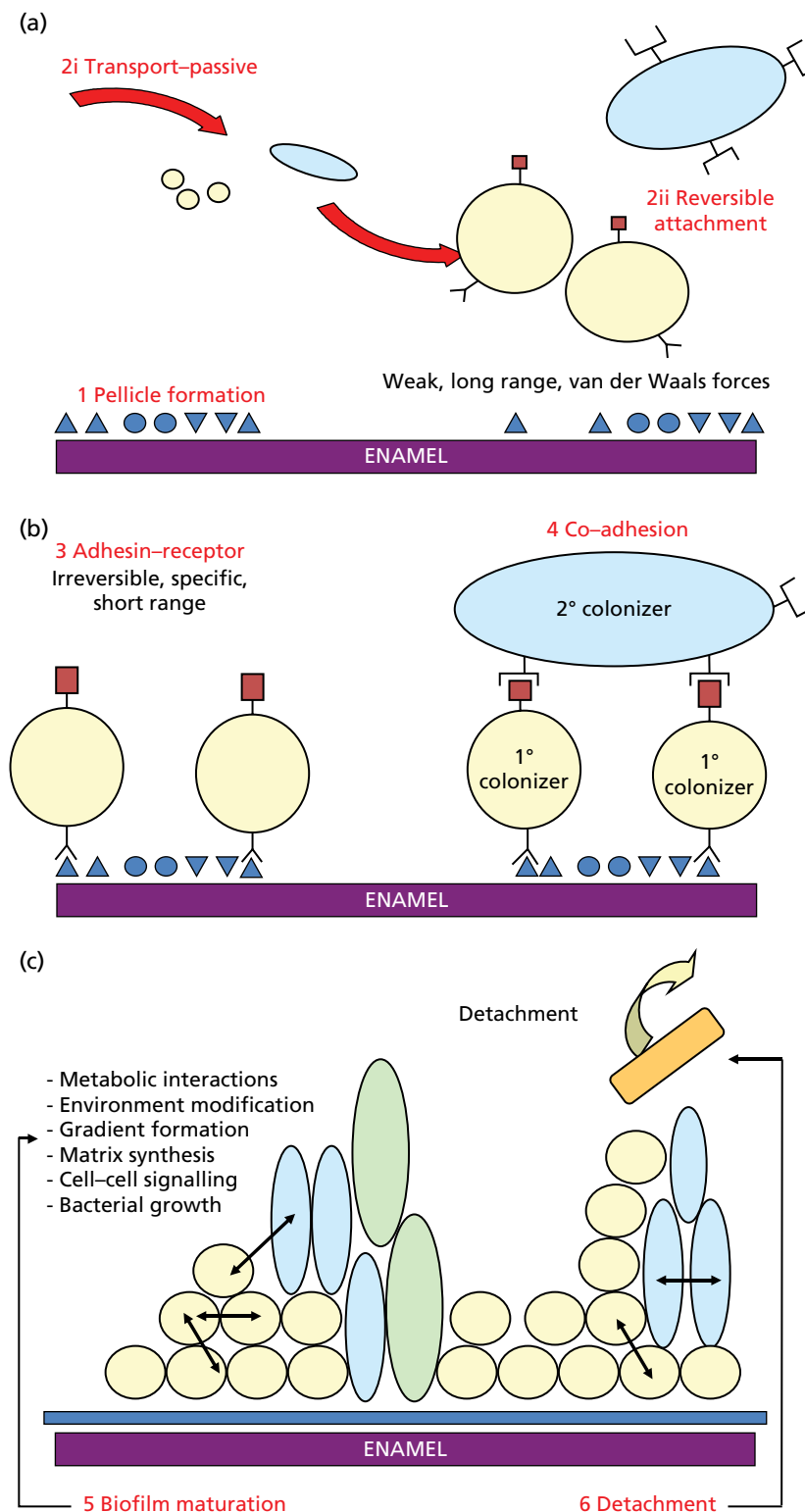


Fig. 8-2 Schematic representation of the different stages in the formation of dental biofilms. (a) Pellicle forms on a clean tooth surface (1). Bacteria are transported passively to the tooth surfaces (2i), where they may be held reversibly by weak, long-range forces of attraction (2ii). (b) Attachment becomes more permanent through specific stereochemical molecular interactions between adhesins on the bacterium and complementary receptors in the pellicle (3), and secondary colonizers attach to the already attached primary colonizers by molecular interactions (co-adhesion) (4). (c) Growth results in biofilm maturation, facilitating a wide range of intermicrobial interactions (synergistic and antagonistic) (5). On occasions, cells can detach to colonize elsewhere (6). (Source: Marsh & Martin 2009. Reproduced with permission from Elsevier.)

- (Kolenbrander *et al.* 2010), leading to an increase in microbial diversity
5. Multiplication of the attached cells, leading to an increase in biomass and synthesis of exopolymers to form the biofilm matrix (plaque maturation)

6. Detachment of attached cells to promote colonization elsewhere.

These stages will now be described in more detail.

Conditioning film formation

Bacteria rarely colonize clean enamel. Within seconds of eruption, or following cleaning, tooth surfaces become coated with a conditioning film of molecules (biologically-active proteins, phosphoproteins, and glycoproteins) derived mainly from saliva (but also from GCF and from the bacteria themselves) (Hannig *et al.* 2005). The conditioning film alters the biologic and chemical properties of the surface, and the composition of the pellicle directly influences the pattern of subsequent microbial colonization. Microorganisms interact directly with this conditioning film (Fig. 8-3).

Reversible and more permanent attachment

Initially, only a limited number of bacterial species are able to attach to the conditioning film. Bacteria can be held reversibly near to the surface by weak, long range, physicochemical forces between the electrical charge of the molecules on the pellicle-coated surface and those on the cell surface (Bos *et al.* 1999). This reversible adhesion creates the opportunity for stronger and more permanent attachment to be established. Molecules (adhesins) on these early bacterial colonizers (mainly streptococci, e.g. *Streptococcus mitis*, *Streptococcus oralis*) can bind to complementary receptors in the acquired pellicle to make the attachment stronger (Busscher *et al.* 2008; Nobbs *et al.* 2011). Individual species can deploy multiple adhesins

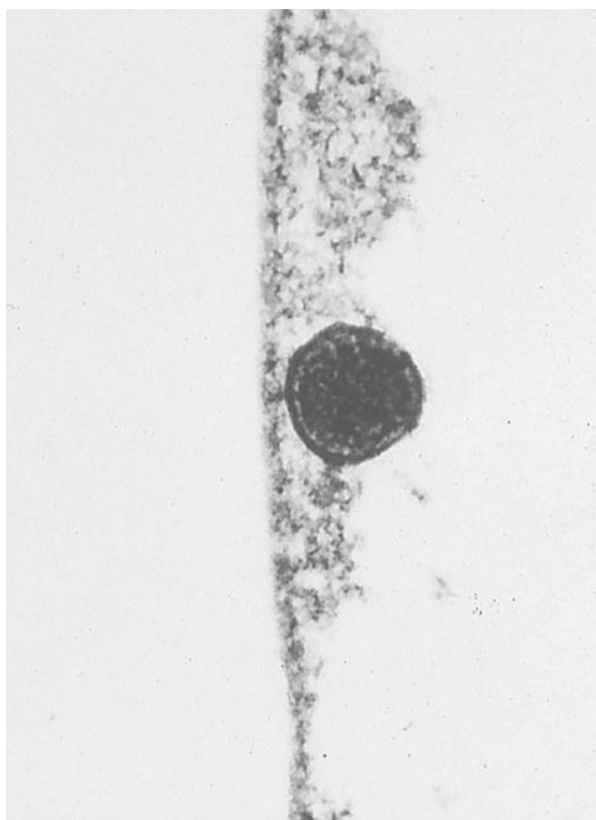


Fig. 8-3 Electron micrograph illustrating a 4-hour pellicle with a single bacterial cell attached. (Source: Brex *et al.* 1981. Reproduced with permission from John Wiley & Sons.)

(Nobbs *et al.* 2011); in Gram-positive bacteria, several families of surface proteins can act as adhesins, including serine-rich repeat, antigen I/II, and pilus families. In Gram-negative bacteria, auto-transporters, extracellular matrix-binding proteins, and pili function as adhesins (Nobbs *et al.* 2011).

Co-adhesion

Once attached, the pioneer colonizers start to multiply. The metabolism of these bacteria that attach early modifies the local environment, for example by making it more anaerobic following their consumption of oxygen and the production of reduced end products of metabolism. As the biofilm develops, adhesins on the cell surface of more fastidious secondary colonizers, such as obligate anaerobes, bind to receptors on bacteria that are already attached by a process termed co-adhesion or co-aggregation, and the composition of the biofilm becomes more diverse (a process termed microbial succession) (Kolenbrander *et al.* 2006) (Figs. 8-4, 8-5). A key organism in plaque biofilm

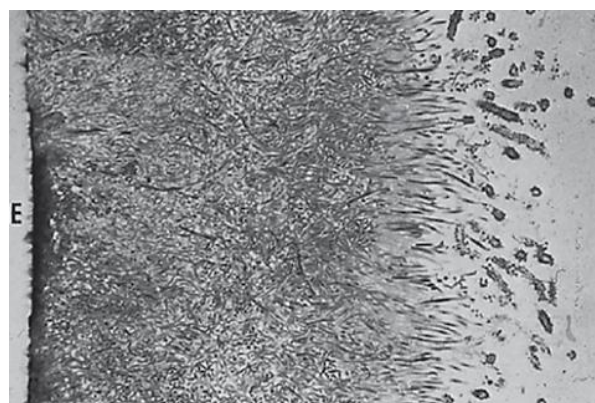


Fig. 8-4 Semi-thin section of a supragingival biofilm on enamel (E) which has been dissolved prior to sectioning. Magnification $\times 750$. (Source: Listgarten 1976. Reproduced with permission from the American Academy of Periodontology.)

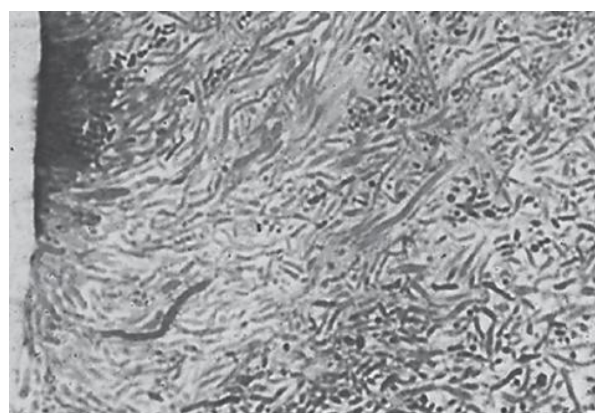


Fig. 8-5 Semi-thin section of supragingival biofilm. Filamentous organisms predominate. At the surface, some of these organisms are surrounded by cocci; the configuration resembles a corn-cob. Magnification $\times 1400$. (Source: Listgarten 1976. Reproduced with permission from the American Academy of Periodontology.)

development is *Fusobacterium nucleatum*. This species can co-adhere to most oral bacteria, and acts as an important bridging organism between early and late colonizing species. Co-adhesion may help ensure that bacteria co-locate with other organisms with complementary metabolic functions.

Plaque maturation

Some of the attached bacteria synthesize extracellular polymers (the plaque matrix) that can consolidate attachment of the biofilm. The matrix is more than a mere scaffold for the biofilm; it can bind and retain molecules, including enzymes, and also retard the penetration of charged molecules into the biofilm (Allison 2003; Vu *et al.* 2009; Marsh *et al.* 2011). Biofilms are spatially- and functionally-organized, and the heterogeneous conditions within the biofilm induce novel patterns of bacterial gene expression, while the close proximity of different species provides the opportunity for interactions (Kuramitsu *et al.* 2007; Hojo *et al.* 2009; Marsh *et al.* 2011). Examples of these interactions include:

- Development of food chains (in which the end product of metabolism of one organism is used as a primary nutrient by secondary feeders) and metabolic cooperation among species to catabolize structurally-complex host macromolecules. These interactions increase the metabolic efficiency of the microbial community (Periasamy & Kolenbrander 2010; Marsh *et al.* 2011).
- Cell-cell signaling. Plaque bacteria have been shown to communicate with one another in a cell density-dependent manner via small diffusible molecules, using strategies similar to those described for other biofilms, for example by the secretion of small peptides by Gram-positive bacteria to coordinate gene expression among cells of a similar species (Suntharalingam & Cvitkovitch 2005). In *Streptococcus mutans*, quorum sensing is mediated by a competence stimulating peptide (CSP) (Li *et al.* 2002). This peptide also induces genetic competence in *S. mutans* so that the transformation frequency of biofilm-grown *S. mutans* was 10–600-fold greater than for planktonic cells. Lysed cells in biofilms could then act as donors of DNA, thereby increasing the opportunity for horizontal gene transfer in dental plaque. This quorum sensing system also functions to regulate acid tolerance in *S. mutans* biofilms. It has been proposed that *S. mutans*, upon exposure to low pH, could release CSP, and initiate a coordinated “protective” response among neighboring cells to this potentially lethal stress.

Other communication systems may function between different oral species (Kolenbrander *et al.* 2002). *LuxS* genes encode for autoinducer-2 (AI-2), and have been detected in several genera of oral

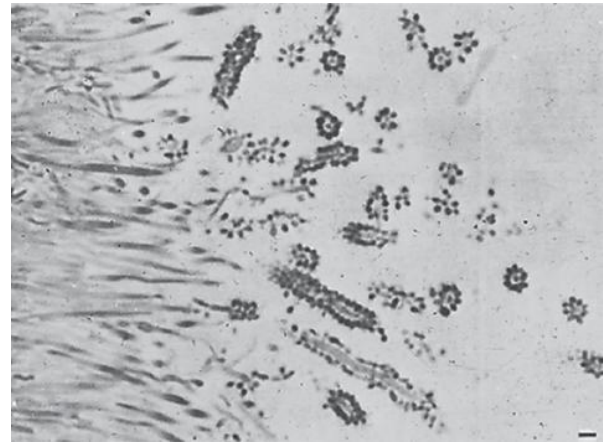


Fig. 8-6 “Corn-cob” formations seen at the biofilm surface shown in Figs. 8-4 and 8-5. Magnification $\times 1300$. Bar 1 μm (Source: Listgarten 1976. Reproduced with permission from the American Academy of Periodontology.)

Gram-positive and Gram-negative bacteria, implying that AI-2 may signal across a broader species range. Several putative periodontal pathogens (*F. nucleatum*, *P. intermedia*, *P. gingivalis*, *Aggregatibacter actinomycetem-comitans*) secrete a signal related to AI-2 (Fong *et al.* 2001; Frias *et al.* 2001).

Characteristic cell associations can be seen in mature dental plaque, such as “corn-cob” (in which coccoid-shaped cells attach along the tip of filamentous organisms; Fig. 8-6) and “test-tube brush” (rod-shaped bacteria sticking out perpendicularly from bacterial filaments) formations (Zijnga *et al.* 2010). These will be discussed further in the next section.

The structure of biofilms facilitates the likelihood of successful horizontal gene transfer. As discussed above, signaling molecules such as CSP markedly increase the ability of recipient cells in biofilms to take up DNA. The transfer of conjugative transposons encoding tetracycline resistance between streptococci has been demonstrated in model biofilms. The recovery from the nasopharynx of resident (*S. mitis*, *S. oralis*) and pathogenic (*S. pneumoniae*) bacteria with penicillin resistance genes showing a common mosaic structure confirms that gene transfer can occur *in vivo* (Dowson *et al.* 1990; Hakenbeck *et al.* 1998). Similar evidence suggests sharing of genes responsible for penicillin-binding proteins among commensal and pathogenic *Neisseria* (Bowler *et al.* 1994).

Structure of dental biofilms

Early studies using electron microscopy (EM) gave important insights into the structure of dental plaque from different sites, and demonstrated the presence of biofilms containing a range of morphologic types of microorganism, often as a compacted structure. More recently, confocal laser scanning microscopy has been used, which does not require the type of sample processing as for EM, and so samples can be seen in their natural, hydrated state. In confocal

microscopy, biofilms are optically sectioned and reconstructed as three-dimensional structures using appropriate software. The use of confocal microscopy has confirmed that dental plaque can also have an open architecture similar to that of biofilms from other habitats. Channels have been observed within biofilms, although they may also contain exopolymers (Wood *et al.*, 2000; Auschill *et al.* 2001). Various microscopy techniques can also be combined with new staining methods to give more detail about the composition and organization of these biofilms. For example, the use of live/dead stains has indicated that bacterial vitality varies throughout the biofilm, with the most viable bacteria present in the central part of plaque, and lining the voids and channels (Auschill *et al.* 2001). This more open architecture should enable molecules to readily move in and out of plaque, but the presence of a matrix comprised of a diverse range of exopolymers creates a complex environment for accurately predicting the penetration and distribution of molecules within plaque (Robinson *et al.* 1997; Thurnheer *et al.* 2003; Marcotte *et al.* 2004), including the delivery of oral care and therapeutic agents. Fluorescent *in situ* hybridization (FISH), in which a fluorescent probe is combined with an oligonucleotide probe specific for a particular organism, can show the location of these microbes, including unculturable species, within the biofilm (Marsh *et al.* 2011; Zijnga *et al.* 2010) (Fig. 8-7).

Supragingival plaque biofilms appear to be quite heterogeneous in structure when viewed using FISH probes designed against a range of bacterial species (Zijnga *et al.* 2010). Subgingival plaque has an even more complex architecture. Initial studies using conventional light microscopy identified distinct tooth-associated and epithelial cell-associated biofilms, with the possibility of a less dense zone of

organisms between the two (Socransky & Haffajee 2002). Co-adhesion is considered to be important in the temporal and spatial development of subgingival biofilms. Many of the putative pathogens that can be found in subgingival dental plaque in health are present in low numbers, and persist by attaching to the early streptococcal and actinomyces colonizers (Kuboniwa & Lamont 2010). These developing complex microbial communities engage in numerous metabolic interactions in order to survive.

Recently, subgingival biofilms were observed directly on extracted teeth, and the identification and location of bacteria was investigated using FISH (Zijnga *et al.* 2010). The architecture of subgingival biofilms was shown to be complex, with four layers being identified. The basal layer was composed of rod-shaped bacteria (*Actinomyces* spp.) attached perpendicularly to the tooth surface, above which was an intermediate layer composed of many spindle-shaped cells, including *F. nucleatum* and *Tannerella forsythia*. In the top layer were many putative periodontal pathogens, such as *P. gingivalis*, *Porphyromonas endodontalis*, *P. intermedia*, and *Parvimonas micra*. A fourth layer of unattached cells mainly consisted of spirochetes. Also, *Synergistetes* spp. formed a palisade-like layer along the outer edge of the biofilm, and were in direct contact with host immune cells. These bacteria are very difficult to grow in pure culture (most are currently “unculturable”), but can make up a sizeable proportion of subgingival plaque, and their location may indicate an important role in modulating host-biofilm interactions (Zijnga *et al.* 2010). The FISH studies allowed the identification of the species involved in the characteristic aggregates seen in subgingival biofilms. Lactobacilli formed the central axis of some of the “test-tube brushes”, with organisms such as *Tannerella*, *F. nucleatum*, and

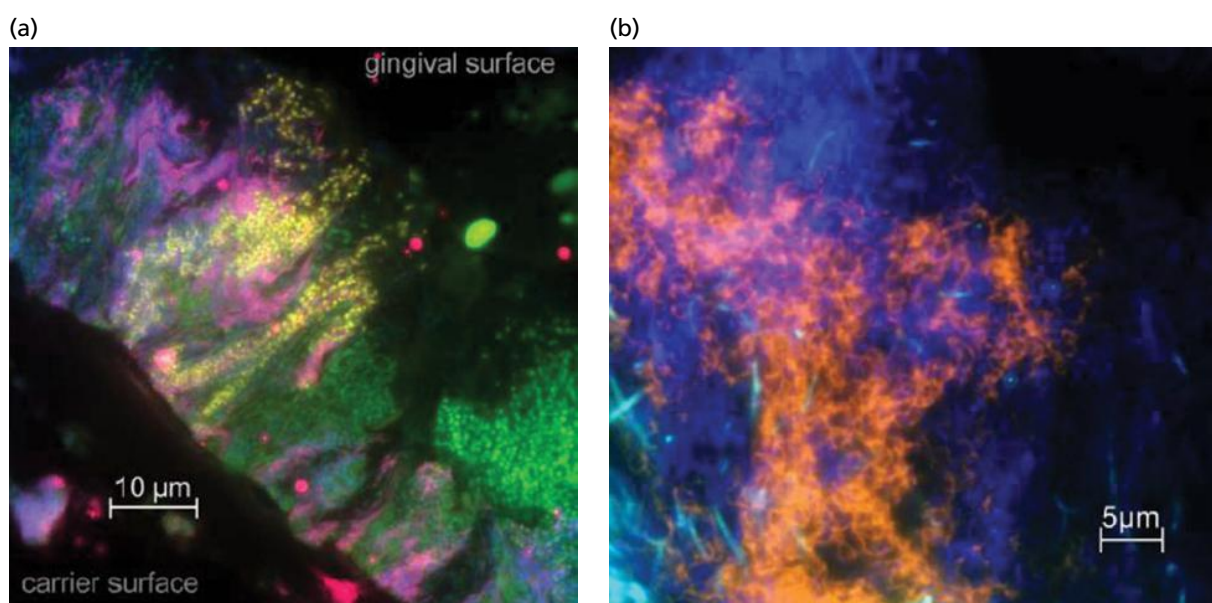


Fig. 8-7 Samples of subgingival biofilms with various oral species detected by fluorescence *in situ* hybridization (FISH). (a) Clusters of *Fusobacterium* spp. (magenta) and *Prevotella intermedia* (yellow). (b) Unculturable spirochetes (orange) and *Fusobacterium* spp. (light blue). (Source: Marsh & Moter 2011a. Reproduced with permission from John Wiley & Sons.)

Synergistetes spp. radiating from this central cell. Corn-cobs consisting of streptococci adhering to a central axis of yeast cells or hyphae were observed by FISH (Zijnga *et al.* 2010). Corn-cobs formed between streptococci and *Corynebacterium matruchotii*, and between *Veillonella* spp. and *Eubacterium* spp. have also been reported.

Bacterial metabolism in plaque results in the development of gradients within dental biofilms in parameters that are critical to microbial growth (nutrients, pH, oxygen, etc.). These gradients are not necessarily linear; the use of two-photon excitation microscopy coupled with fluorescent lifetime imaging demonstrated considerable heterogeneity in pH over relatively short distances in model mixed culture oral biofilms (Vroom *et al.* 1999). Such environmental heterogeneity will allow fastidious bacteria to survive in plaque, and enable microorganisms to coexist that would be incompatible with one another in a more homogeneous environment. This explains how organisms with apparently contradictory metabolic and growth needs (e.g. in terms of atmospheric and nutritional requirements) are able to persist at the same site.

Microbial composition of dental biofilms

As discussed earlier, the mouth supports the growth of a characteristic microbiota which includes viruses, mycoplasma, bacteria, Archaea, fungi, and protozoa (Marsh & Martin 2009). Bacteria are the most numerous group and, initially, they were characterized using traditional cultural approaches. Over time, it became apparent that there was a large discrepancy between the number of bacteria in a sample that could be grown by these conventional approaches and those that were observed directly by microscopy (Choi *et al.* 1994; Paster *et al.* 2001). It is estimated that only about 50% of the resident oral microbiota can currently be cultivated in pure culture in the laboratory (Wade 1999, 2002). This may be due to our ignorance of the growth requirements of some species, but probably also reflects our naivety in attempting to isolate microbes in pure culture that have evolved over millennia to grow with other species as part of a community (Vartoukian *et al.* 2010a, b).

Our knowledge of the richness and diversity of the resident oral microbiota has been enhanced by the recent application of culture-independent, molecular approaches (Pozhitkov *et al.* 2011; Wade 2011). The accumulated data from numerous studies of different surfaces and sites based around amplification, cloning, and sequencing of the 16S rRNA gene have identified around 900 species in the mouth. Most sites (mucosal or plaque) yielded 20–30 different predominant species, while the number of species per individual mouth can range from 34 to 72 (Aas *et al.* 2005). However, these figures may still be an underestimate and the use of more powerful, high throughput,

next-generation sequencing platforms will better detect low abundance species.

Underway is the “Human Oral Microbiome” project, which aims to identify and characterize all members of the resident oral microbiota found in health and disease (Dewhirst *et al.* 2010). Data are being placed in the publically accessible web-based Human Oral Microbiome Database (<http://www.homd.org>), which also feeds information into the larger Human Microbiome Project. It is beyond the scope of this chapter to describe the properties of members of the resident oral microbiota, and the reader is recommended to refer to this website or to other specialist texts for more detail (Marsh & Martin 2009).

Biofilms develop on mucosal and dental surfaces, and the microbial composition of the biofilm varies at distinct sites on a tooth (fissures, approximal surfaces, gingival crevices), reflecting the inherent differences in their anatomy and biology (Fig. 8-8) (Aas *et al.* 2005; Sachdeo *et al.* 2008; Marsh & Martin 2009; Papaioannou *et al.* 2009). The properties of each habitat will select for those organisms that are best adapted to the prevailing conditions, and are able to persist. This also means that any changes to the environment will directly impact on the composition and activity of the biofilm. The remainder of this chapter will focus on the properties of dental biofilms.

The normal microbiota of *fissures* is relatively sparse and the organisms present have a saccharolytic metabolism (i.e. their energy is derived from sugar catabolism), and are either aerobic or facultatively anaerobic. The predominant bacteria are streptococci, many of which produce extracellular polysaccharides, and there are few Gram-negative or anaerobic organisms (Theilade *et al.* 1982). The properties of this site are strongly influenced by saliva.

In contrast, the *gingival crevice* has a more diverse microbiota, including many Gram-negative anaerobic and proteolytic species; this is due to the lower Eh at this site and the delivery of a distinct set of proteins and glycoproteins by the GCF (Slots 1977). The black pigmented anaerobes have an absolute requirement for hemin for growth, and these organisms can obtain this co-factor from the degradation of heme-containing host molecules present in GCF. Molecular studies using culture-independent approaches have emphasized the fact that the gingival crevice supports the most diverse microbial communities in the healthy mouth, with 40% of the amplified clones representing novel phylotypes. Many currently unculturable species can be detected at this site (Figs. 8-7b, 8-8).

Approximal surfaces have a microbiota that is intermediate in composition between that of fissures and gingival crevices, and also harbors many anaerobic species. These sites have high proportions of *Actinomyces* spp. (Bowden *et al.* 1975).

As discussed earlier, once established, the composition of the resident microbiota at any site remains relatively stable over time, unless there are marked changes to the habitat. Importantly, this stability,

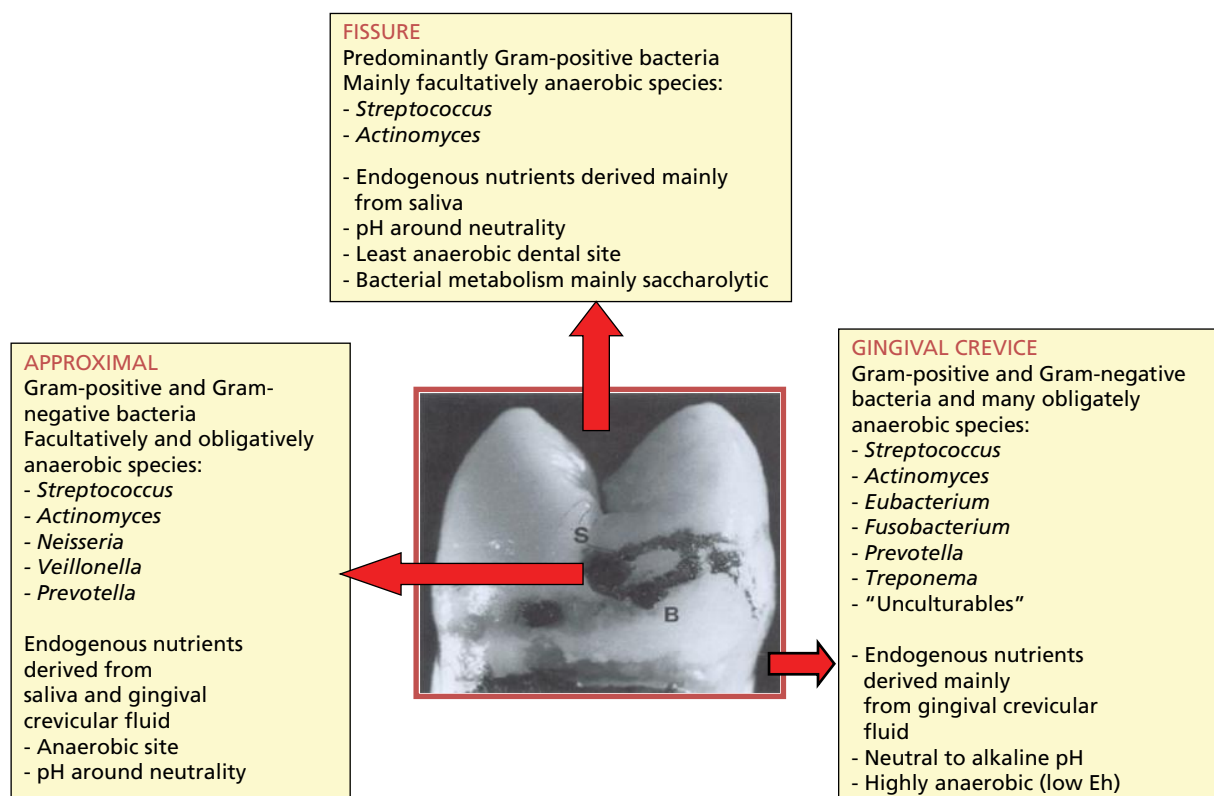


Fig. 8-8 Predominant groups of bacteria found at distinct sites on the tooth surface, and key features of each habitat.

termed microbial homeostasis, stems not from any metabolic indifference by the resident microbiota, but reflects a highly dynamic state in which the relative proportions of individual species are held in balance due to the numerous interactions, both synergistic and antagonistic, described earlier (Marsh 1989). This natural balance is maintained despite continual surveillance by the host defenses and the regular exposure of the mouth to a variety of modest environmental stresses, such as via the diet, changes in saliva flow, and oral hygiene (see Fig. 8-1a). However, microbial homeostasis can break down on occasions if one of the key parameters affecting growth is perturbed and this perturbation is sufficiently robust or regular to result in the reorganization of the composition of the biofilm, with the outgrowth of previously minor components (see Fig. 8-1b). Such perturbations can be due to immunologic (e.g. neutrophil dysfunction, immune suppression, etc.) or non-immunologic (e.g. xerostomia, diet change, etc.) factors, and can predispose a site to disease (Marsh & Martin 2009; Marsh *et al.* 2011b), and forms the basis of the "ecological plaque hypothesis" (Marsh 2003) that describes the dynamic relationship between the oral microbiota and the host in health and disease.

Benefits to the host of a resident oral microbiota

The host has a sophisticated array of host defenses provided by both the innate and adaptive arms of the immune system, the primary function of which is to protect tissues against microbial colonization and

invasion. Despite these host defenses, the host has evolved over millennia to support a complex resident microbiota and, at first sight, this might appear paradoxical (the "commensal paradox") (Henderson & Wilson 1998). It is now apparent that the resident microbiota confers considerable benefit to the host, and that these natural microbial residents are essential for the normal development of the physiology, nutrition, and defenses of the host (Marsh 2000; Wilks 2007) (Fig. 8-9).

The biologic mechanisms that permit a constructive coexistence between the host and the resident microbiota, while enabling the host to retain the capacity to respond to exogenous microbial insults, are now being dissected. The host is not indifferent to the presence of the diverse microbial communities that reside on its surfaces. It is actively engaged in cross-talk with its resident microbiota in order to effectively maintain a constructive relationship. The host is able to detect microorganisms, and has evolved systems to enable it to tolerate resident microorganisms without initiating a damaging inflammatory response, while also being able to mount an efficient defense against pathogens. Pathogenic and non-pathogenic bacteria may initiate different intracellular signaling pathways and innate immune responses in epithelial cells (Canny & McCormick 2008; Hooper 2009; Neish 2009). Certain oral streptococci have been shown to suppress epithelial cell cytokine expression (Hasegawa *et al.* 2007; Peyret-Lacombe *et al.* 2009). *Streptococcus salivarius* K12 not only down-regulated epithelial cell inflammatory responses by inhibiting the NF- κ B pathway,

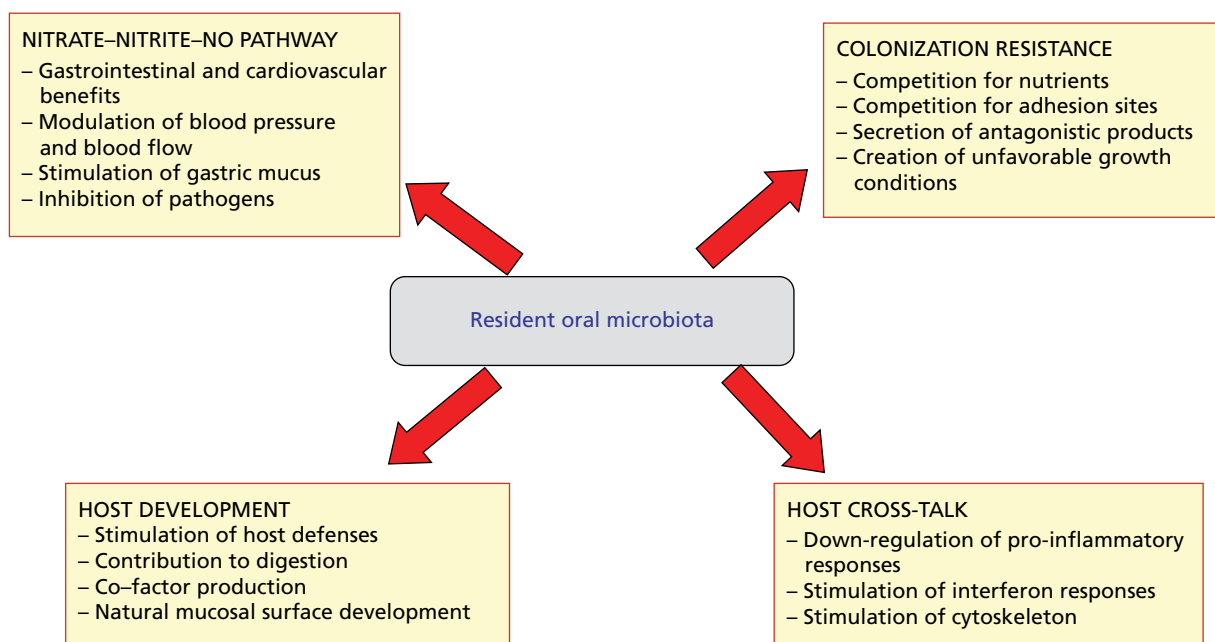


Fig. 8-9 Beneficial functions of the resident oral microbiota.

but also actively stimulated beneficial pathways, including type I and II interferon responses, and exerted significant effects on the cytoskeleton and adhesive properties of the host cell (Cosseau *et al.* 2008). The “commensal communism” paradigm proposes that our oral microbiota and mucosa form a unified “tissue” in which host-microbe “cross-talk” is finely balanced to ensure microbial survival and prevent the induction of damaging inflammation (Henderson & Wilson 1998).

One of the principal benefits emanating from the existence of a resident microbiota at a site is the ability to prevent colonization by exogenous (and often pathogenic) microorganisms. This property, termed “colonization resistance” (Van der Waaij *et al.* 1971), is due to various properties of resident microbes, including more effective (1) attachment to host receptors, (2) competition for endogenous nutrients, (3) creation of unfavorable growth conditions to discourage attachment and multiplication of invading organisms, and (4) production of antagonistic substances (hydrogen peroxide, bacteriocins, etc.). Colonization resistance can be impaired by factors that compromise the integrity of the host defenses or perturb the stability of the resident microbiota, such as the side effects of cytotoxic therapy or the long-term use of broad-spectrum antibiotics (Johnston & Bodley 1972). For example, the latter can suppress the resident bacterial oral microbiota, permitting overgrowth by previously minor populations of oral yeasts. Attempts to boost colonization resistance using replacement therapy (in which resident organisms are deliberately re-implanted), for example after periodontal therapy (Teughels *et al.* 2007) or by the use of probiotics (Devine & Marsh 2009), are being explored, although the evidence for benefits from oral probiotics is still equivocal.

The resident oral bacteria play an important role in maintaining many important aspects of the gastrointestinal and cardiovascular systems, via the metabolism of dietary nitrate. Approximately 25% of ingested nitrate is secreted in saliva where facultatively anaerobic oral resident bacteria reduce nitrate to nitrite. Nitrite affects a number of key physiologic processes, including the regulation of blood flow, blood pressure, gastric integrity, and tissue protection against ischemic injury. Nitrite can be further converted to nitric oxide in the acidified stomach. This has antimicrobial properties and contributes to defense against enteropathogens and in the regulation of gastric mucosal blood flow and mucus formation. Significantly, studies have shown that the use of an antimicrobial mouthwash (Govoni *et al.* 2008; Petersson *et al.* 2009) or a broad-spectrum antibiotic (Dougall *et al.* 1995) reduced the microbial conversion of nitrate to nitrite with a loss of the biologic benefits of nitrite, including reduced gastric mucus thickness and loss of an expected fall in blood pressure.

Concluding remarks

The mouth supports the establishment of diverse communities of microorganisms. These communities, and those present at other habitats in the body, play an active and critical role in the normal development of the host and in the maintenance of health. Clinicians need to be aware of the beneficial functions of the resident oral microbiota, so that treatment strategies are focused on the control rather than the elimination of these natural biofilms. In conclusion, oral care practices should attempt to maintain plaque at levels compatible with health in order to retain the beneficial properties of the resident oral microbiota, while preventing microbial excesses that increase the risk of dental diseases.

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